

RESPONSE TO REVIEWERS

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1. RESPONSE FOR REVIEWER 4140.

We thank the reviewer for the positive assessment of the novelty of our work and for the constructive comments. Below, we clarify the main concerns and explain how we will further improve the manuscript for clarity.

1. Technical Claims: Clarify imprecise statements (e.g., "learning GFT bases"). Detail the actual implementation/complexity of $\sigma(\mathbf{R}\mathbf{R}^\top) \odot \mathbf{A}$.

The phrase "learning GFT bases" may be imprecise and we thank the reviewer for pointing this out. GRASP does not directly compute or parameterize eigenvectors of the graph Laplacian. Instead, it learns two subgraph structures, namely homophilic and heterophilic subgraphs, through graph structure learning (GSL). Each learned graph induces its own Laplacian operator and the corresponding spectral space. These induced spectral spaces naturally capture smooth (low-frequency) and non-smooth (high-frequency) signals, respectively. Based on them, GRASP applies low-pass and high-pass filters on the corresponding subgraphs. This design avoids frequency mixing that often arises when all signals are filtered on a single fixed graph spectrum, and helps prevent suboptimal embeddings. In the revised manuscript, we will replace the phrase "learning GFT bases" with a more accurate description that emphasizes learning graph structures that induce different spectral properties.

Regarding the complexity of $\sigma(\mathbf{R}\mathbf{R}^\top) \odot \mathbf{A}$, the discussion in the current manuscript refers to space complexity. It is $O(cN + M)$, where $c \ll N$, N is the number of nodes, and M is the number of edges. The time complexity is $O(cN^2 + M)$, which is dominated by forming $\mathbf{R}\mathbf{R}^\top$, while $\mathbf{A}$ remains sparse. In the revision, we will clearly distinguish between space and time complexity to avoid confusion.

2. Experiments: Strengthen validation by: (a) using cleaned/newer heterophily benchmarks (e.g., Roman-Empire); (b) adding comparisons to node-wise adaptive spectral methods (e.g., NODE-MoE, DSF); (c) providing diagnostics on the learned subgraphs (e.g., edge homophily, spectral concentration).

To address these concerns, we report additional experimental results in Table 1. We include newer heterophilic benchmarks such as Roman-Empire and compare GRASP with recent node-wise adaptive spectral methods, including DSF and

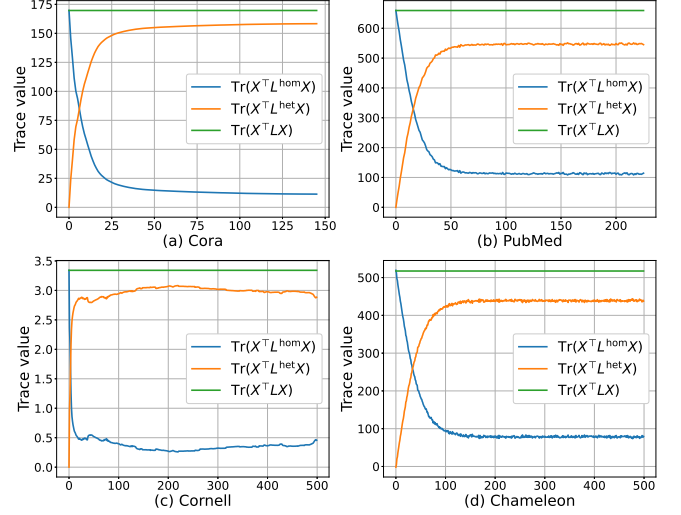


Fig. 1: The behavior of embedding smoothness and roughness regularizers during training on Cora, PubMed, Cornell and Chameleon. The x-axis stands for the training epochs.

NODE-MoE. Since NODE-MoE is not publicly released, its results are reported as stated in the original paper. Across both homophilic and heterophilic datasets, GRASP consistently outperforms the compared baselines. On Roman-Empire, GRASP achieves 85.59% accuracy, compared to 75.18% for DSF, indicating strong robustness under heterophily.

We also provide diagnostic analyses of the learned subgraphs using spectral smoothness and roughness measured by $\text{Trace}(\mathbf{X}^\top \mathbf{L} \mathbf{X})$, as shown in Fig. 1. On Chameleon, $\text{Trace}(\mathbf{X}^\top \mathbf{L}_{\text{hom}} \mathbf{X})$ decreases from 520 to 110 during training, indicating increasingly smooth signals on the learned homophilic subgraph. In contrast, $\text{Trace}(\mathbf{X}^\top \mathbf{L}_{\text{het}} \mathbf{X})$ increases from near zero to 410, showing that signals on the heterophilic subgraph become more non-smooth. This divergence supports that GRASP separates low- and high-frequency components through graph splitting.

3. Theory: Substantiate the "tighter bound" claim with a clearer comparison to prior work or an empirical proxy.

The reviewer's concern regarding the "tighter bound" mainly relates to presentation clarity. In [1], prediction difference is upper-bounded by a term proportional to $\|h_u - h_v\| + \rho|e_u - e_v|$, where e_u denotes local homophily. Under heterophily, the second term may dominate due to unstable

Table 1: Node classification accuracy. We highlight the best results in bold and the second-best results by underlining.

Dataset	Homophilic			Heterophilic				
	Cora	Citeseer	PubMed	Chameleon	Cornell	Texas	Squirrel	Roman-Empire
GCN	87.78 $\pm$ 0.96	81.39 $\pm$ 1.23	88.90 $\pm$ 0.32	64.18 $\pm$ 2.62	82.46 $\pm$ 3.11	83.11 $\pm$ 3.20	44.76 $\pm$ 1.39	73.69 $\pm$ 0.74
GraphSAGE	86.58 $\pm$ 0.26	78.24 $\pm$ 0.30	86.85 $\pm$ 0.11	62.15 $\pm$ 0.42	71.41 $\pm$ 1.24	79.03 $\pm$ 1.20	41.26 $\pm$ 0.26	85.74 $\pm$ 0.67
GOAL	88.75 $\pm$ 0.87	77.15 $\pm$ 0.95	89.25 $\pm$ 0.55	71.65 $\pm$ 1.66	84.90 $\pm$ 0.66	92.02 $\pm$ 0.74	60.53 $\pm$ 1.60	72.30 $\pm$ 0.48
FAGCN	<u>88.85 $\pm$ 1.36</u>	<u>82.37 $\pm$ 1.46</u>	89.98 $\pm$ 0.54	65.47 $\pm$ 2.84	88.03 $\pm$ 5.60	88.85 $\pm$ 4.39	42.24 $\pm$ 1.20	65.22 $\pm$ 0.56
GPR-GNN	79.51 $\pm$ 0.36	67.63 $\pm$ 0.38	85.07 $\pm$ 0.09	67.48 $\pm$ 0.40	91.36 $\pm$ 0.70	92.92 $\pm$ 0.61	49.93 $\pm$ 0.53	64.85 $\pm$ 0.27
BernNet	88.52 $\pm$ 0.95	80.09 $\pm$ 0.79	88.48 $\pm$ 0.41	68.29 $\pm$ 1.58	92.13 $\pm$ 1.64	93.12 $\pm$ 0.65	51.35 $\pm$ 0.73	65.56 $\pm$ 1.34
UniFilter	88.28 $\pm$ 1.36	77.98 $\pm$ 5.28	<u>91.24 $\pm$ 0.69</u>	<u>73.63 $\pm$ 2.03</u>	85.74 $\pm$ 3.36	89.02 $\pm$ 3.52	65.75 $\pm$ 0.96	74.43 $\pm$ 0.25
DSF	89.63 $\pm$ 0.17	78.22 $\pm$ 0.29	90.51 $\pm$ 0.07	71.64 $\pm$ 0.55	84.93 $\pm$ 0.90	85.56 $\pm$ 0.93	58.44 $\pm$ 0.30	75.18 $\pm$ 0.37
NODE-MoE	89.38 $\pm$ 1.26	77.78 $\pm$ 1.36	89.58 $\pm$ 0.60	73.64 $\pm$ 1.80	-	-	62.31 $\pm$ 1.98	-
GRASP	90.77 $\pm$ 0.30	82.80 $\pm$ 0.43	91.40 $\pm$ 0.55	73.78 $\pm$ 2.27	93.82 $\pm$ 1.20	94.08 $\pm$ 2.64	<u>64.75 $\pm$ 1.71</u>	<u>85.59 $\pm$ 0.41</u>

neighborhood compositions. In contrast, our bound in Theorem 1 removes the dependency on $|e_u - e_v|$ by constructing homophilic and heterophilic subgraphs. This leads to a tighter bound in heterophilic settings. We will revise the manuscript to make this comparison clearer.

4. Clarity: Define ablation settings precisely (e.g., "w/o rel", " L_{F-hom} ") and clarify hyperparameter details (e.g., τ , layer-wise sharing).

We will revise the manuscript to clearly define all ablation settings, such as "w/o rel" and L_{F-hom} , and to clarify hyperparameter details, including τ , which is shared across layers.

We believe these revisions fully address the reviewer’s concerns, improve the clarity and presentation of the paper, and further demonstrate the soundness of our approach.

2. RESPONSE FOR REVIEWER 8E71.

We respectfully clarify several points that may stem from a misunderstanding of our work, or because the reviewer focused primarily on the introduction. The questions raised are largely limited to high-level background and motivation, which suggests that some well-established assumptions and standard interpretations in heterophilic GNN research may not have been fully considered.

1. The paper would be strengthened by providing concrete examples of real-world networks that exhibit heterophily where connected nodes are likely to have different labels/attributes to better motivate the practical significance of the problem setting.

Heterophily has been studied for several years in graph representation learning, and the observation that connected nodes may belong to different classes or exhibit dissimilar attributes has become a common premise in this line of research [2, 3, 4]. They have provided concrete real-world examples to motivate this setting, including dating networks, where linked nodes are typically of different genders, and molecular graphs, where connected atoms often belong to different types. These examples illustrate that heterophily naturally arises in many practical applications.

In our paper, all datasets are real-world benchmarks that are widely used in heterophilic GNN research. Specifically, Chameleon and Squirrel [5] are Wikipedia-based networks, where nodes correspond to web pages on different topics and edges represent hyperlinks between pages. Node features are extracted from informative nouns in the text, and labels indicate page traffic levels. Cornell and Texas [6] are subsets of the WebKB dataset, where nodes represent university web pages and edges denote mutual hyperlinks. In these datasets, neighboring nodes frequently belong to different semantic categories, which provides a practical and well-established motivation for studying heterophilic graphs.

2. The statement that “traditional GNNs often produce unsatisfactory representations on heterophilic graphs” is currently asserted without sufficient support. Please add appropriate citations to prior work discussing this issue and/or include brief experimental evidence (even a small reference to baseline results) to substantiate the claim.

The statement that “traditional GNNs often produce unsatisfactory representations on heterophilic graphs” is not a new claim introduced by our paper, but a commonly accepted assumption in heterophilic GNN research [7, 8, 9]. These works consistently show that classical message-passing models, such as GCN and GraphSAGE, rely on neighborhood aggregation mechanisms that emphasize feature similarity and smoothness. While this behavior is effective under homophily, it becomes problematic when connected nodes are likely to belong to different classes or have dissimilar attributes.

Our experimental results are fully consistent with this established understanding. As shown in Table 1, GCN and GraphSAGE exhibit clear performance degradation on heterophilic benchmarks compared with methods designed to handle heterophily. We will make this connection more visible in the introduction by adding citations and pointing readers to the corresponding experimental results, without treating this observation as a new or debatable claim.

3. The manuscript would benefit from a clearer and more detailed explanation of what “low-frequency” and “high-frequency” information correspond to in het-

erophilic graphs. In particular, please clarify the intuition behind why high-frequency components become informative under heterophily, why separating low/high frequencies is important for node classification, and how these concepts relate to smoothing vs. preserving inter-class differences.

The interpretation of low- and high-frequency information adopted in this paper follows standard definitions in spectral GNN literature [5, 4]. Low-frequency components correspond to smooth signals over the graph, while high-frequency components capture rapid variations that are often aligned with inter-class differences in heterophilic graphs. This perspective is already introduced in Section 2 (Preliminaries) and further reflected in the design of our model in Section 3 (Methodology). To facilitate understanding for readers who may be less familiar with spectral views of heterophily, we will add clarifying remarks that connect these standard definitions to the heterophilic setting.

4. There are still some typos and minor language issues in the manuscript (e.g., “ghe” → “the”). A careful proofreading is recommended to improve readability and presentation quality

We will carefully proofread the manuscript and correct the remaining typos and minor language issues in the revised version.

3. REFERENCES

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